



AWS - Monitoring

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Topics Covered

CloudWatch

Alarms

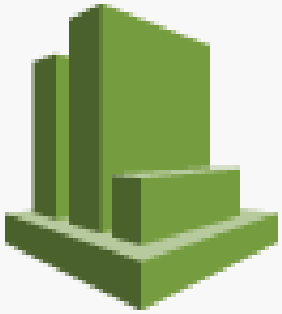
SNS

Events

Cloud Trail

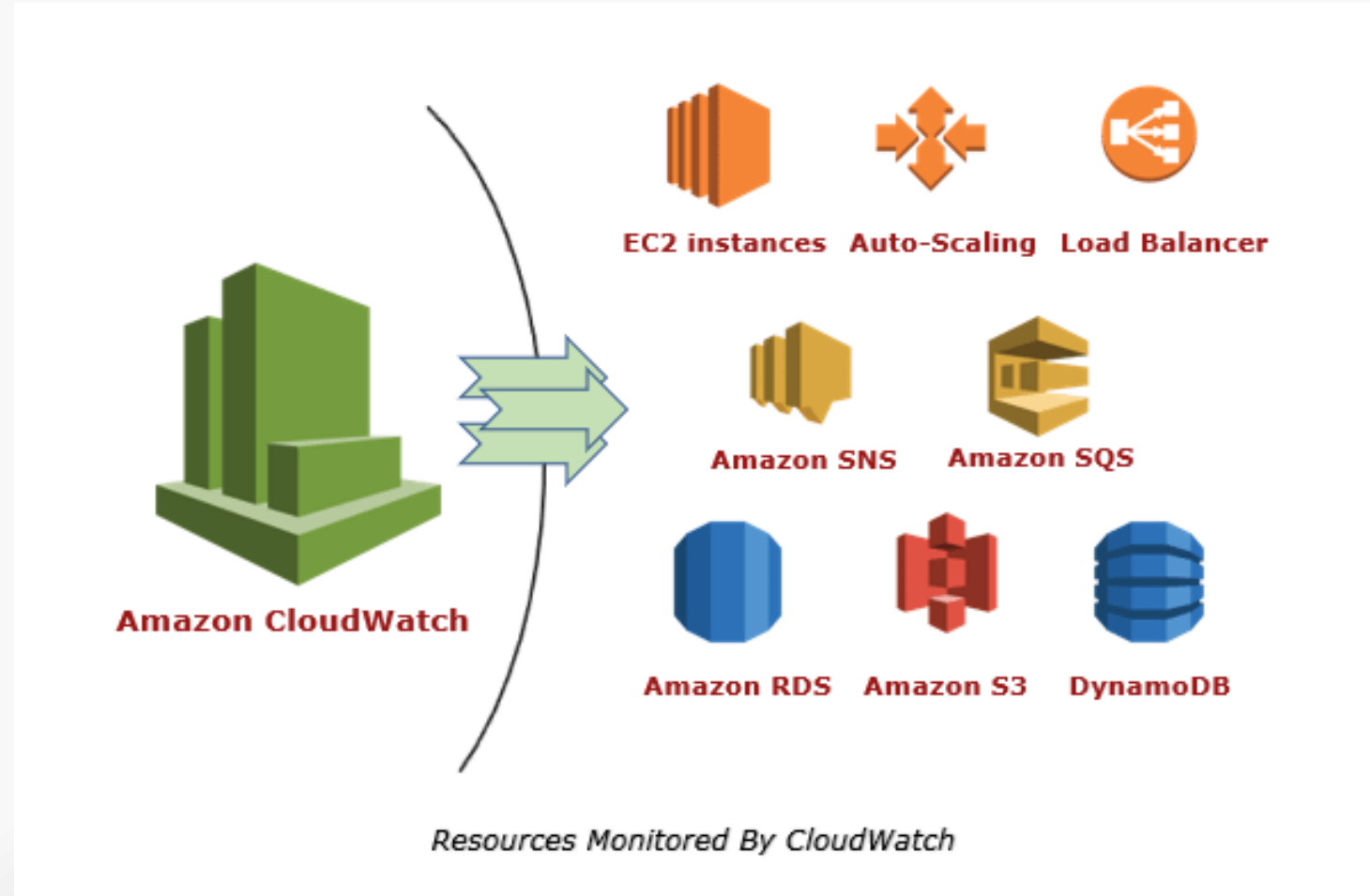


AWS - CloudWatch



What is AWS CloudWatch ??

- *Amazon CloudWatch is the component of Amazon Web Services that provides real-time monitoring of AWS resources and customer applications running on Amazon infrastructure.*



Concepts of CloudWatch

- Collects Monitored data for AWS components like.
 - Virtual instances hosted in **Amazon EC2**
 - Databases located in **Amazon RDS**
 - Data stored in **Amazon S3**
 - Elastic Load Balancer**
 - Auto-Scaling Groups**
 - Many Other resources
- Monitors, stores and provides access to system and application log files
- There are default Values for the Monitoring “Period” for every component that are free, NOT charged
 - Eg: -- EC2 is monitored every 5 min, S3 is monitored every 1 day

Concepts of CloudWatch – Continued...

- Provides various alert capabilities, including rules and triggers high resolutions alarms and sends notifications
- Collects and provides a real-time presentation of operational data in form of key metrics like CPU utilization, disk storage etc.
- By default the monitored DATA is saved for “15 months”.

AWS CloudWatch Parameters

Metrics

- ***Metrics*** represents a time-ordered set of data points that are published to CloudWatch
- Metrics are uniquely defined by a name, a namespace, and zero or more dimensions
- Each data point has a time-stamp

Dimensions

- A ***dimension*** is a name/value pair that uniquely identifies a metric
- Dimensions can be considered as categories of characteristics that describe a metric.

AWS CloudWatch Parameters

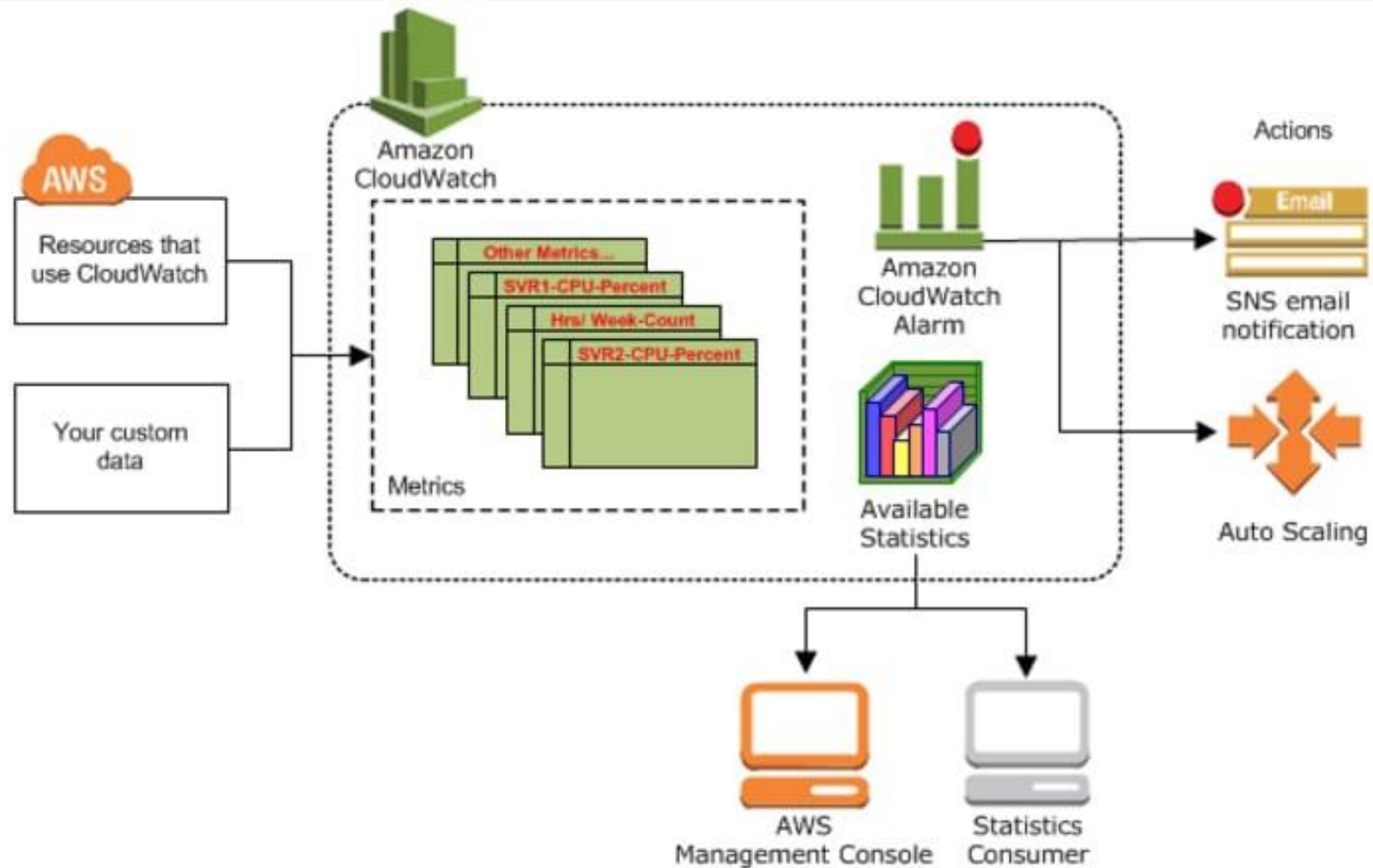
Statistics

- ***Statistics*** are metric data aggregations over specified periods of time
- Aggregations are made using the namespace, metric name, dimensions within the time period you specify.
- *Few available statistics are maximum, minimum, sum, average and sample count*

Alarm

- An ***alarm*** can be used to automatically initiate actions on your behalf
- It watches a single metric over a specified time period and performs one or more specified actions
- The action is simply a notification that is sent to Amazon SNS topic.

Working of AWS CloudWatch



Default Metrics

- EC2
 - CPU, Disk, Network, StatusCheck
- S3 and Glacier
 - Bucketsizebytes, NumberofObjets

AWS - Alarms



Alarm

- Alarm has 3 States
 - **OK**
 - **INSUFFICIENT DATA**
 - **ALARM**

Alarm

- Alarm is created to take an ACTION for a particular “Situation” or usage of the component.
- Type of Action, eg:-- Alarm on EC2
 - Notification
 - Autoscaling
 - EC2 action

Alarm Types for an EC2 Instance

Create Alarm

You can use CloudWatch alarms to be notified automatically whenever metric data reaches a level you define.

To edit an alarm, first choose whom to notify and then define when the notification should be sent.

☒ **Send a notification to:** CPU-High (vishwaawsac@gmail.com) [create topic](#)

☒ **Take the action:**

- ☐ Recover this instance [i](#)
- ☐ Stop this instance [i](#)
- ☐ Terminate this instance [i](#)
- ☐ Reboot this instance [i](#)

Whenever: Average ▾ of CPU Utilization ▾

Is: >= ▾ Percent

For at least: 1 consecutive period(s) of 5 Minutes ▾

Name of alarm:

[Cancel](#) [Create Alarm](#)

CPU Utilization Percent

i-0538dd0472edd6315

Alarm types for Load Balancer

Create Alarm

You can use CloudWatch alarms to be notified automatically whenever metric data reaches a level you define.

To edit an alarm, first choose whom to notify and then define when the notification should be sent.

☒ **Send a notification to:** CPU-High (vishwaawsac@gmail.com) [create topic](#)

Whenever: Average of

Is: >=

For at least: 1 consecutive

Name of alarm: awsnetworkelb-net-l

Active flow count

Active flow count

New flow count

ConsumedLBCapacityUnits

Load balancer reset count

Client reset count

Target reset count

Processed Bytes

Active flow count Count

net/LB01/c

AWS – SNS – Simple Notification Service



AWS - CloudTrail

